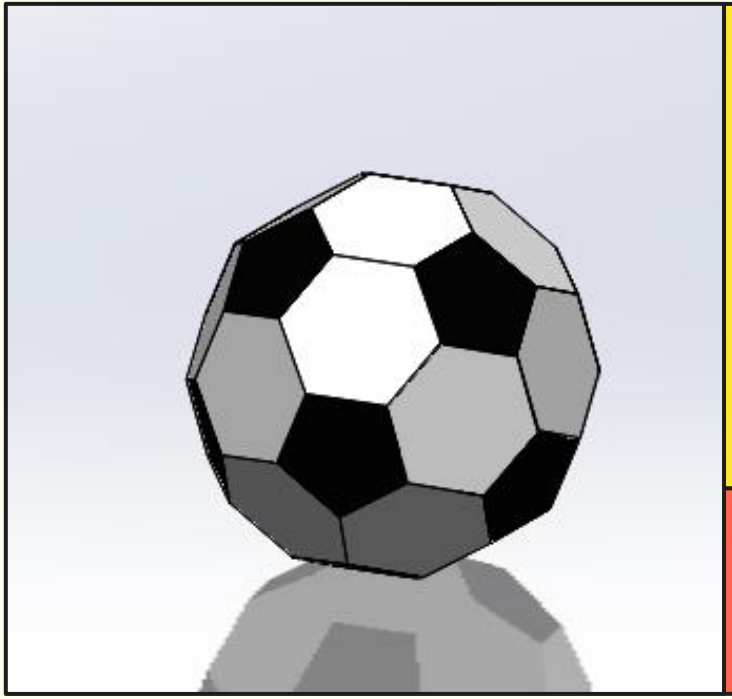
10/23/26
Friday
MAE 1001

Solidworks Crash Course 2!

Solidworks Assembly Essentials!

Let's Recap



We made 1/70th of a Soccer Ball!

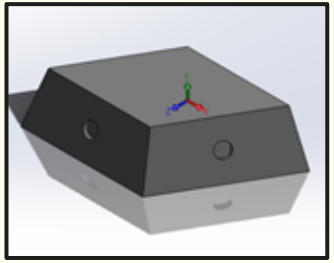
Today, we're going to introduce:

- Adding a Part to an **Assembly**
- Basic **Mating** in an Assembly
- Finite Element Analysis and what it does?

These three skills will help you put you in a great position early your **professional career!**

But first! Does anyone want to share their customized Part?

Your Base Part



Remember your Customization Criteria

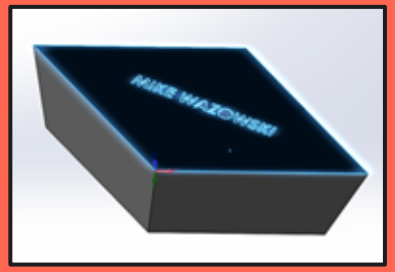
No Extrusions over 2cm!

Let's try to conserve our materials the best we can and not be wasteful while still having fun!



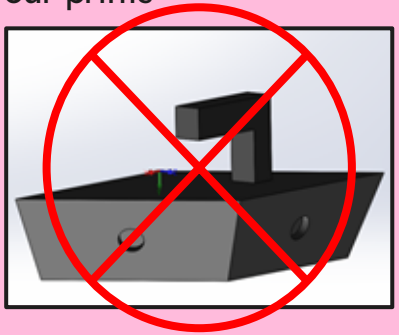
Your name must be on it!

Use the A (Text) tool to add your name to the drawing and either cut or extrude it out ~0.2 cm



No Overhangs!

We are trying to print with **NO SUPPORTS**, we can do this as long as we have no overhangs on our prints

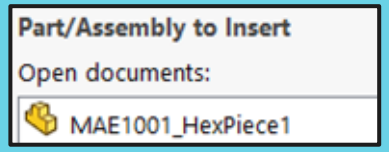


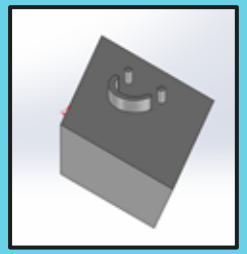
Have fun and be original!

Remember, this is supposed to be fun! Try something new, if it doesn't work, ask me, or a friend, or the internet.



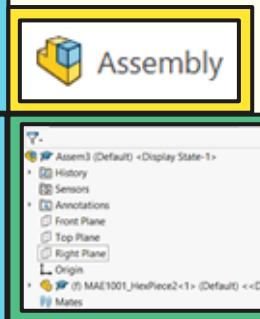
Starting an Assembly

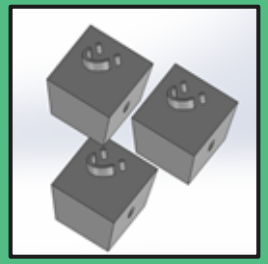




Open a Part in an Assembly

First, open the assembly tab seen at the top of the slide, then select your part. Click and place the part anywhere on your work area.

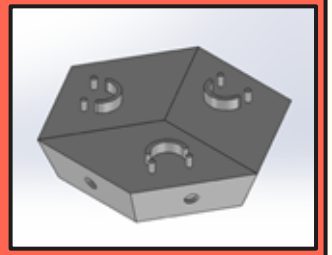




Let's Make 3!

Highlight, then copy and paste the first part added into the assembly two more times





Forming a Full Hexagon

Mating the three parts together can be very simple by mating the parts in a manner that keeps the parts from moving. Use the 

mate type and select faces/points/vertices that should be touching to accomplish this.

Assembliception

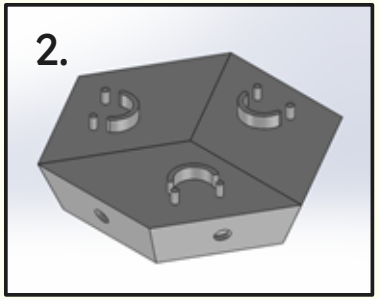


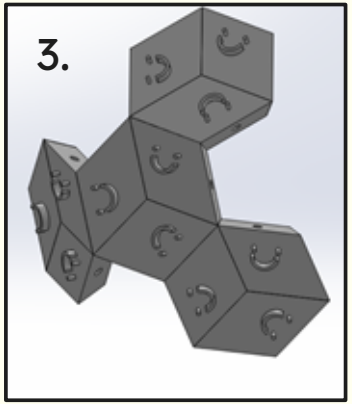
An **Assembly** can be made of **Assemblies**.

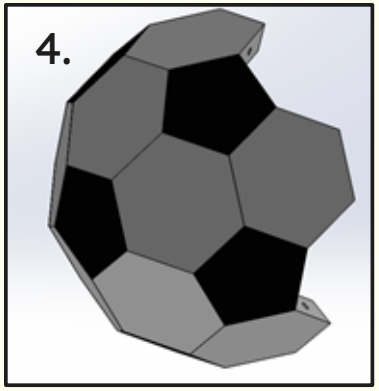
1. Start a **New Assembly**
1. Add your custom **Hexagon Assembly**
1. **Duplicate it** three times and mate them to the sides

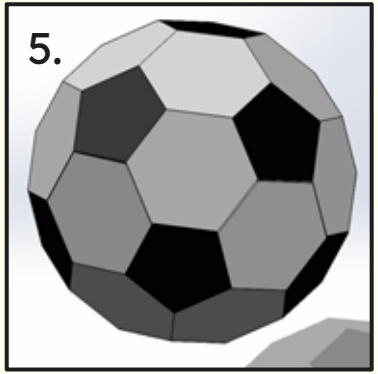
How'd we get to the soccer ball?

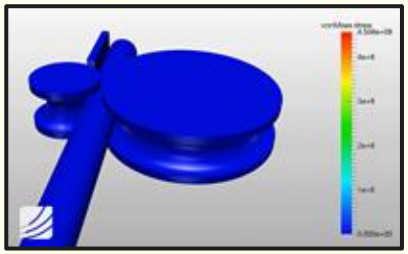
1. Adding pentagons into the assembly, you can create a half soccer ball
1. Creating an assembly of two half soccer balls results in the full ball!

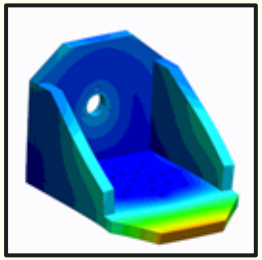


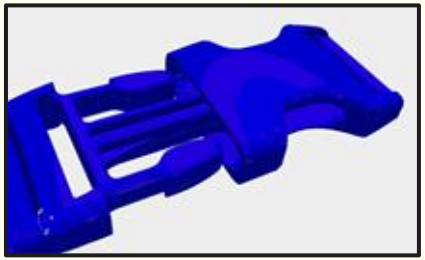












Briefly: What is FEA?

Finite Element Analysis

"A **computerized** method used to predict how a **product** reacts to real-world **forces** such as **stress, vibration, heat, and fluid flow**. It helps engineers and manufacturers understand whether a product will **break, wear out, or function** as designed." (Autodesk)

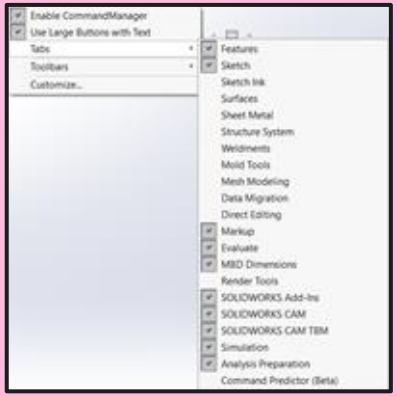
You too can do FEA!

Setting up a Simulation:

1. First, open up your **original Hex Part**

2. Then use the **find** or **activate** the Simulation tab

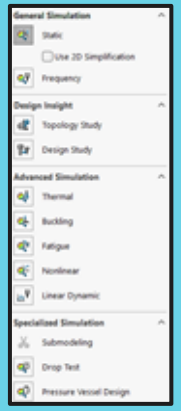
To see what tabs that you have active, **right click** on the tabs and check:



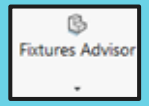
Start a Simulation

1. Select 

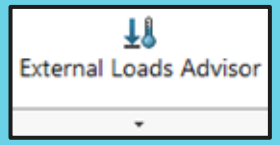
2. Read through the different **types of simulation** and then select the **green check**:



3. Apply a **material**: 

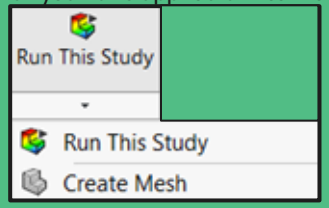
4. **Fixture** the part (how is it being held?): 

5. Add a **load** on the part:

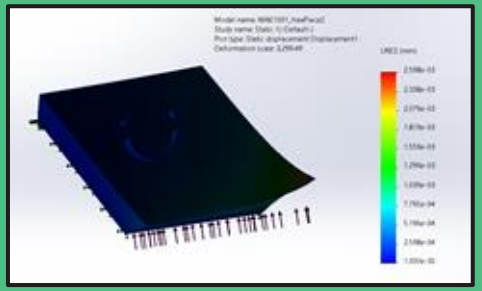


Run the Simulation!

1. Make sure that you have applied a mesh to the part:



2. Run the Study!



Remember: Keep in mind your scale!

Q&A

